

Assumptions Log

This document records all assumptions made during the EMS Readiness Optimization project.

Data Assumptions

A1: Crash Data Representativeness

Assumption: Historical crash data (2013-2026) is representative of future crash patterns.

Rationale: Long time series provides stable estimates of spatial and temporal patterns.

Risk: Major changes in traffic patterns (e.g., congestion pricing) could alter future demand.

Mitigation: Include sensitivity analysis on demand levels.

A2: Location Accuracy

Assumption: Crash latitude/longitude coordinates are sufficiently accurate for demand estimation.

Rationale: NYPD geocoding process should provide reasonable accuracy for most incidents.

Risk: Some records may have missing or inaccurate coordinates.

Mitigation: Filter records with missing coordinates; aggregate to grid cells.

A3: Firehouse Availability

Assumption: All Manhattan firehouses are available as potential EMS staging locations.

Rationale: Study explores optimal staging locations; operational constraints addressed separately.

Risk: Some firehouses may not have space or infrastructure for EMS staging.

Mitigation: Document as limitation; can add constraints in future iterations.

Model Assumptions

A4: Travel Time Model

Assumption: Travel times can be approximated using Euclidean distance with a scaling factor or simplified network distances.

Rationale: Full routing adds complexity without fundamentally changing optimization insights.

Risk: May over/underestimate actual response times in some areas.

Mitigation: Sensitivity analysis on travel time multipliers.

A5: Single Unit Response

Assumption: Each crash incident requires exactly one EMS unit.

Rationale: Simplifies simulation while capturing core dispatch dynamics.

Risk: Major incidents may require multiple units.

Mitigation: Model captures most incidents; note as limitation.

A6: Homogeneous Service Time

Assumption: Service time (on-scene + transport) follows a single distribution.

Rationale: Data on actual service times not available; use literature estimates.

Risk: Service times may vary by incident severity.

Mitigation: Sensitivity analysis on service time parameters.

A7: Immediate Dispatch

Assumption: EMS units are dispatched immediately upon incident occurrence.

Rationale: Focuses on travel time rather than dispatch delay.

Risk: Real systems have dispatch processing time.

Mitigation: Can add dispatch delay parameter if needed.

Geographic Assumptions

A8: Manhattan Boundary

Assumption: Incidents and firehouses are classified by whether they fall within Manhattan borough boundary.

Rationale: Standard geographic classification for NYC.

Risk: Boundary edges may have some ambiguity.

Mitigation: Use official borough boundary polygons.

A9: CBD Definition

Assumption: CBD is defined as the MTA Congestion Relief Zone boundary.

Rationale: Provides consistent, official boundary for the high-density core.

Risk: Other CBD definitions exist; results depend on boundary choice.

Mitigation: Document specific boundary used; can test alternatives.

Operational Assumptions

A10: Unlimited Staging Capacity

Assumption: Any number of EMS units can stage at a single firehouse.

Rationale: Simplifies optimization; capacity constraints are secondary.

Risk: Physical constraints may limit actual staging.

Mitigation: Document as limitation; add capacity constraints in extensions.

A11: No Pre-emption

Assumption: Once an EMS unit is assigned to an incident, it completes service before taking new calls.

Rationale: Standard EMS operational policy.

Risk: Pre-emption for high-priority calls does occur.

Mitigation: Aligns with most common practice; note limitation.

A12: Manhattan Distance Approximation

Assumption: Manhattan (taxicab) distance provides an upper-bound approximation of actual road distance in a grid-based network; Haversine provides a lower bound.

Rationale: Manhattan's street grid is largely rectangular; true road distance falls between Haversine and Manhattan distance.

Risk: Diagonal avenues and non-grid areas (e.g., lower Manhattan) deviate from both approximations.

Mitigation: Compared both metrics side-by-side; P2 allocation is robust to metric choice (mean RT differs by <0.01 min in simulation).

A13: CBD-Focused Demand Weighting

Assumption: Applying a 3× demand weight to CBD precincts in the CBD-focused model is a reasonable multiplier to test CBD prioritisation.

Rationale: The 3× factor substantially up-weights CBD demand (from 56% to ~77% effective weight) without completely ignoring non-CBD.

Risk: A different multiplier (e.g., 2× or 5×) might yield different conclusions.

Mitigation: Results show that even with 3× weight, CBD RT does not improve (2.50 vs. 2.47 min), while non-CBD RT degrades dramatically (+159%). This confirms that the Manhattan-wide P2 is robust and any CBD-focused up-weighting is counterproductive.

Assumption Review Schedule

Assumption	Review Date	Reviewer	Status
All	Initial	Project Lead	Documented

Last Updated: March 12, 2026